

KYUHYEOK SEO

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EDUCATION

Pennsylvania State University (PSU)

Ph.D. in Informatics

Advisor: James Z. Wang

University Park, PA, USA

Aug 2025 - May 2030

Seoul National University (SNU)

B.S. in Industrial Engineering and Computer Science & Engineering

GPA 3.69/4.00 (Cum Laude), 3.75/4.00 (IE), 3.89/4.00 (CSE)

Seoul, Korea

Mar 2018 - Aug 2024

Gyeonggi Science High School for the Gifted (GSHS)

Diploma for Gifted Students in Math & Science

Suwon, Korea

Mar 2015 - Feb 2018

EXPERIENCE

Research Assistant

College of Information Science and Technology, PSU

University Park, PA, USA

Aug 2025 - Present

- Assistant to Professor James Z. Wang in developing multi-modal deep learning models for cross-modal learning

Ads Performance, Data Science & Analytics (DSA) Intern

MOLOCO, Inc.

Seoul, Korea

Jun 2024 - Sep 2024

- Redefined and validated ad error criteria by analyzing their impact on performance decline
- Developed Looker Studio dashboards to visualize advertising KPIs across various dimensions
- Automated regular dashboard updates using Google BigQuery and Airflow

Undergraduate Researcher

Data Mining Lab (DM) at KAIST

Seoul, Korea

Jan 2023 - Dec 2023

- Developed a deep learning framework for Open-World Graph Learning (OWG) by leveraging Graph Contrastive Learning (GCL) techniques under the supervision of Prof. Kijung Shin
- Improved embedding grouping using extended GCL pairing and novel cluster parameters

Undergraduate Researcher

Data Science & Artificial Intelligence Lab (DSAIL) at KAIST

Daejeon, Korea

Jun 2022 - Aug 2022

- Implemented foundational GNNs and recommender systems, advised by Prof. Chanyoung Park
- Developed regularization strategies to mitigate catastrophic forgetting in multi-task learning

Visiting Student Researcher

Iowa State University

Ames, IA, USA

Oct 2016 - Feb 2017

- Contributed to the publication of the research paper "Enumerating Multiplex Juggling Patterns."

PROJECTS

Bitcoin Volatility prediction

Learning From Data (406.561), SNU

Mar 2024 - Jun 2024

- Engineered an end-to-end pipeline using a Transformer model to forecast BTC volatility, covering data collection, analysis, and evaluation

SKILLS

Programming	Python, SQL (Advanced); C/C++, Java, Verilog (Proficient)
Technologies	PyTorch, Git, LaTeX, AWS, Google BigQuery, Airflow, Looker Studio
Languages	Korean (Native), English (Fluent, TOEFL 102)

EXTRACURRICULAR ACTIVITIES

Auxiliary Police, Korean National Police Agency	Jul 2020 - Jan 2022
Managed daily precinct operations and emergency dispatch duties as part of South Korea’s mandatory military service	

AWARDS AND SCHOLARSHIPS

Merit-based Scholarship, College of Engineering, SNU	Feb 2023
Full scholarship awarded based on grades	
Merit-based Scholarship, College of Engineering, SNU	Aug 2022
Half scholarship awarded based on grades	
SQL Professional (SQLP), Korea Data Industry Agency (K-DATA)	Jun 2022
Certified for expertise in SQL development, data modeling, and performance optimization	